

Setting up your computer

NSDF-Data student guide 1 — Windows edition

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Tip

Who this is for. You are starting from little or no computing experience. You will paste some commands into a terminal and read the answers. You do **not** need to know any programming. Work through the steps in order. Each step ends with a **Checkpoint**: if it does not match, stop there.

Before you start

- **Time:** about one hour. Most of it is waiting for downloads and installers.
- **You need:** an internet connection, the password for your computer, and about 3 GB of free disk space.
- **Rule of thumb:** type or paste commands **exactly** as written. Capitals and spaces matter.
- **Nothing to be afraid of:** the commands in this guide install software and make new folders. None of them deletes your files. If you are not sure what a line does, there is a short table after each group of commands.
- **When something goes wrong:** do not guess. Copy the message, note the step you were on, and ask the project lead. See *If something goes wrong* at the end.

Words you will meet

Word	What it means
Terminal	A window where you type commands and the computer answers in text.
Command	One line you type in the terminal, followed by Enter (or Return).
Folder (or directory)	A place on your computer that holds files and other folders.
Path	The address of a folder, such as C:\Users\YourName\Research.
Git	A program that downloads and keeps track of versions of code.
Repository (repo)	A project folder that Git tracks. Here, NSDF-Data.

Word	What it means
Python	The programming language the analysis is written in. You will not need to write Python to finish these guides.
Conda / Miniforge	A tool that installs Python and packages in separate, named <i>environments</i> .
Environment	A sealed set of programs and versions, so this project's software does not clash with others.
Package	A piece of ready-made software (for example numpy).
Jupyter notebook	A document that mixes text, code and plots, run one piece (<i>cell</i>) at a time.
Branch	A separate line of work in Git, so your changes do not touch the main code.

Step 1. Open a terminal

Click the **Start** button, type Command Prompt, and press **Enter**. A black window with white text opens. This is the terminal you will use for Steps 1 to 3. Miniforge (Step 3) adds a second terminal, the **Miniforge Prompt**, and from Step 4 on you must use that one instead: only it knows the conda command.

In both windows, **right-click** the window to paste. Ctrl+V often does not work there.

Checkpoint

You see a line of text ending in a symbol such as >, \$ or %. Type `echo hello` and press Enter: the word hello should appear under it.

Step 2. Install Git

Go to git-scm.com/download/win. The download of the installer starts by itself; if it does not, click the link on that page to download it. Run the installer. Keep every default setting: click **Next** until you reach **Install**, then click **Install**.

When it finishes, **close the Command Prompt and open a new one** (the old window does not know about Git yet).

Checkpoint

In the terminal, type `git --version` and press Enter. You should see a line starting with git version.

Step 3. Install Miniforge

Miniforge installs Python and the other packages the analysis needs. Its download file is **Miniforge3-Windows-x86_64.exe**.

Download the installer with this link: [Miniforge3-Windows-x86_64.exe](#). It is about 150 MB, so it can

take a few minutes. If your browser asks whether to keep the file, choose **Keep**. (If the link ever stops working, the downloads are also listed on the [Miniforge page](#).) Double-click the downloaded file and follow the installer. Click **Next** (and **I Agree** on the licence page) until you reach these choices:

- When it asks who to install for, choose **Just Me**.
- Keep the default install folder.
- On the **Advanced Installation Options** screen, set the four boxes exactly as in the table. They are the installer's defaults, so you may not need to change anything, but check each one:

Box	Set it to
Create shortcuts (supported packages only)	Ticked. This creates the <i>Miniforge Prompt</i> you will use from Step 4.
Add installation to my PATH environment variable	Unticked. The installer says “not recommended”, and the Miniforge Prompt works without it.
Register Miniforge3 as my default Python	Unticked.
Clear the package cache upon completion	Unticked. (Ticking it also works. It only saves a little disk space.)

Then click **Install**, wait until it finishes, and click **Finish**.

Now close the Command Prompt. Click **Start**, type *Miniforge Prompt*, and press **Enter**. Use this window for the rest of the guide.

Checkpoint

Open a **Miniforge Prompt** (click **Start**, type *Miniforge Prompt*, press **Enter**) and type `conda --version`, then press **Enter**. You should see `conda` followed by a version number.

Careful

Windows may show a blue warning box when you run an installer (“Windows protected your PC”). Click **More info**, then **Run anyway**, only if you downloaded the file from the link in this guide.

Step 4. Download the NSDF-Data code

Tip

What is %USERPROFILE%? It is a shortcut that Windows understands. It stands for your own personal folder on this computer, the one named after you (for example `C:\Users\YourName`). Type it **exactly as written**, with the two percent signs. Do not replace it with your name: Windows fills it in for you.

Your project will live in a folder called *Research*. Create it, go into it, and download the code. Type each line and press **Enter**:

```
mkdir %USERPROFILE%\Research
cd %USERPROFILE%\Research
git clone https://github.com/villano-lab/NSDF-Data.git
```

```
cd %USERPROFILE%\Research\NSDF-Data
```

Downloading with `git clone` does not need a GitHub account. It makes a full copy of the project in a folder named NSDF-Data.

What these lines do

Line starts with	What it does
<code>mkdir</code>	Makes a new folder called Research.
<code>cd</code>	Moves you into a folder. <code>cd</code> stands for <i>change directory</i> , which is the same as opening a folder in a file window.
<code>git clone</code>	Downloads a copy of the project from the internet into a new folder called NSDF-Data.

Checkpoint

Type `dir` and press Enter. You should see NSDF-Data in the list.

Step 5. Create your analysis environment

An environment is a sealed box of programs with fixed versions. We make one for this project, called `darkmatter_cli_env`. Type these lines one at a time, pressing Enter after each. The third line takes several minutes and prints a lot of text: wait for the prompt to come back.

```
conda create -n darkmatter_cli_env python=3.10 -y
conda activate darkmatter_cli_env
python -m pip install nsdf-dark-matter==0.3.0 nsdf-dark-matter-cli==0.5.0
python -m pip install numpy==2.2.6 matplotlib==3.10.7 h5py pytest ipykernel
python -m pip install jupyterlab
```

What these lines do

Line	What it does
<code>conda create</code>	Makes a new, empty environment called <code>darkmatter_cli_env</code> , with Python 3.10 inside it.
<code>conda activate</code>	Switches that environment on. While it is on, your prompt starts with <code>(darkmatter_cli_env)</code> .
<code>pip install nsdf-dark-matter...</code>	Installs the software that reads the detector data.
<code>pip install numpy...</code>	Installs software for numbers, plots and checking that everything works.
<code>pip install jupyterlab</code>	Installs Jupyter, the notebook program you will use later.

Tip

The project folder also has a file called `requirements.txt` that lists the same software versions. You do **not** need it: the lines above do the whole job, and they work the same on every computer.

Tip

Every time you open a new terminal, run `conda activate darkmatter_cli_env` first. When it is active, the start of each line shows `(darkmatter_cli_env)`.

Checkpoint

The last line of the output is a message, not an error, and the `(darkmatter_cli_env)` label is at the start of your prompt.

Step 6. Check that everything works

First, the tests. They use made-up data, so they need no download. Type each line and press Enter:

```
conda activate darkmatter_cli_env
cd %USERPROFILE%\Research\NSDF-Data\python
python -m pytest -q
```

Checkpoint

The last line says `52 passed` (it can be a larger number later, which is also fine). If it says failed, stop and ask the project lead, and do not change any files.

Then, the data tool:

```
nsdf-cli version
```

Checkpoint

It prints `NSDF Dark Matter CLI: 0.5.0` (or a later version).

Finally, the notebook program:

```
jupyter lab --version
```

Checkpoint

It prints a version number, such as `4.4.0`. If it says `jupyter-lab is not found`, run `python -m pip install jupyterlab` and try again.

What these commands do

Line	What it does
<code>conda activate</code>	Switches your environment on. Do this in every new terminal.
<code>cd</code>	Moves into the <code>python</code> folder of the project.

Line	What it does
<code>python -m pytest -q</code>	Runs the project's built-in self-checks. The <code>-q</code> just means "keep the output short". 52 passed means every check worked.
<code>nsdf-cli version, jupyter lab --version</code>	Each one only asks a program to say which version it is. If it answers, it is installed.

Step 7. Let Jupyter notebooks use your environment

Jupyter is the notebook program you will use. Register your environment with it once:

```
python -m ipykernel install --user --name darkmatter_cli_env
```

Checkpoint

It prints a line that starts with `Installed kernelspec darkmatter_cli_env`.

Step 8. Get permission to save your work

You will not change the main project. You will save your work on your own **branch**, and for that the lead must give you access:

1. Make a free account at github.com if you do not have one.
2. Send your GitHub username to the project lead. They will add you to the project as a collaborator.
3. Once you have accepted the invitation, tell Git your name and email, so your work is labelled as yours. Use your own details:

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

4. Create your personal branch (your own space to save work, separate from the main project). Use your own name, without spaces, in place of yourname:

```
cd %USERPROFILE%\Research\NSDF-Data
git checkout -b student-yourname
```

Checkpoint

The terminal shows `Switched to a new branch 'student-yourname'`.

Rules for keeping your work safe

- **Never work on master.** That is the shared main version. Work only on your student - branch.
- **Never commit data.** The raw files in the `idx` folder are large and not part of the project. Download only from your home folder, never from inside `NSDF-Data`.
- **Do not run `07221203_2025_dump1_noise.ipynb`** unless the project lead says it is fine. It overwrites a shared archive file.
- **Save often**, and commit your work when a piece is finished (see the first analysis guide).

You are done when

- ☐ Your terminal opens and `git --version` prints a version.
- ☐ `conda --version` prints a version, in a new terminal.
- ☐ The NSDF-Data folder is in %USERPROFILE%\Research.
- ☐ `python -m pytest -q` in the python folder ends with 52 passed.
- ☐ `nsdf-cli version` prints a version.
- ☐ `jupyter lab --version` prints a version.
- ☐ Your Jupyter kernel `darkmatter_cli_env` is installed.
- ☐ You have a student- branch (after access is granted).

Next: *First analysis* (the guide for Windows).

If something goes wrong

1. Read the **last line** of the message. It usually says what is wrong.
2. Close the terminal, open a new one, and run `conda activate darkmatter_cli_env` again. Many problems are just the wrong environment.
3. Copy the command you ran and the message it printed, and send them to the project lead. A screenshot is also fine.

Where to get help

Anthony Villano, project lead: anthony.villano@ucdenver.edu

When you write, include the step number, the command you typed, and the last few lines the computer printed. A screenshot helps. Nobody will be annoyed by a question about a step that did not work.

Session Info

Guide version 7, 7 October 2026, Windows edition. Written for beginners; please tell the project lead where a step was unclear, so the guide can be fixed.

Key links

- Notes site: <https://villano-lab.github.io/NSDF-Data/>
- Repository: <https://github.com/villano-lab/NSDF-Data>
- Miniforge downloads: <https://github.com/conda-forge/miniforge>
- Git downloads: <https://git-scm.com>